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$$\text{force} \cdot \text{displacement}$$

Suppose the force is not constant, and is instead a function $F(x)$ of the position x. Then the work equals

$$\int_a^b F(x)\,dx$$

where a and b are start and end values of x.

Example: I hold a 2 lb ball 3 ft above the ground. I throw the ball upward with a constant 12 lb of force, and it leaves my hand at 5 ft above the ground.

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Suppose instead the force is given by the equation $F(y) = y + 7$, where y is the height above the ground.

A 100 ft long, 200 lb rope hangs from a cliff. How much work is needed to pull the rope halfway up the cliff?

5. (12 points) A 50 ft rope weighing 2 lb/ft is lifting a 5 lb bucket intially carrying 8 lb of water. It is carrying the water up a well that is also 50 ft. The bucket of water is being lifted up at a rate of 3 ft/sec. The bucket has a hole on the bottom, causing water to leak out at a rate of 1/4 lb/sec. Find the work required to lift the rope and leaky bucket of water up to the top of the well.

   [Hint: find the total weight $F(y)$ in pounds of the rope and leaky bucket as a function of distance y from the bottom of the well.]

Average value of a function (6.5)

Let f be a function defined on $[a,b]$. The **average value** of f on $[a,b]$ is

$$f_{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

What is the average value of $\sin x$ on $[0, 2\pi]$?

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The average velocity of a car from $t=0$ s to $t=10$ s is 80 ft/s. What is the car's displacement in this time interval?

The average velocity of a car from $t=0$ s to $t=10$ s is 80 ft/s. Is it true that at some point, the car is moving at exactly 80 ft/s?

Mean Value Theorem for integrals: If f is continuous on $[a,b]$, then there exists a number c is $[a,b]$ such that

$$f(c) = f_{avg} = \frac{1}{b-a}\int_a^b f(x)dx$$